

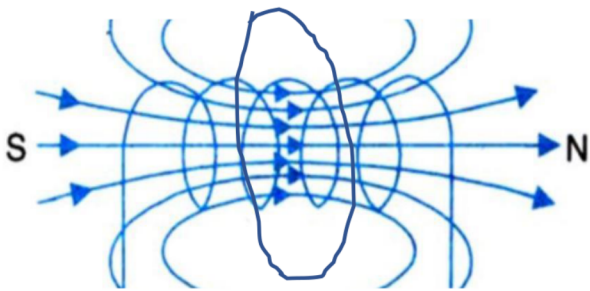
Faraday's Law of Induction & Lenz's Law

Oersted's law

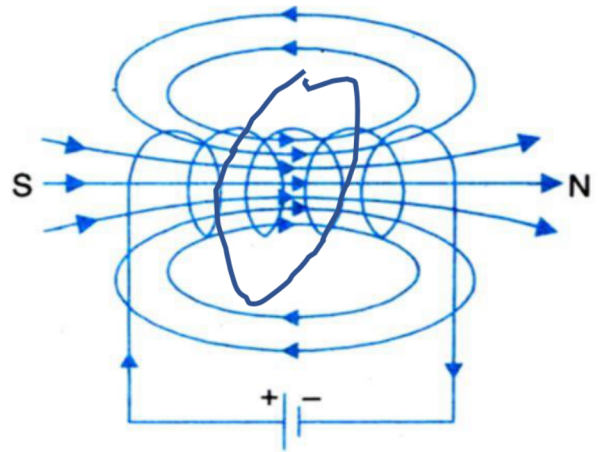
- 1820 by Danish physicist Hans Christian Oersted state that a static current produce static magnetic field.
- This findings is the first connection between Electricity and Magnetism

What Faraday try to understand from Oersted's Law?

Faraday's Observations

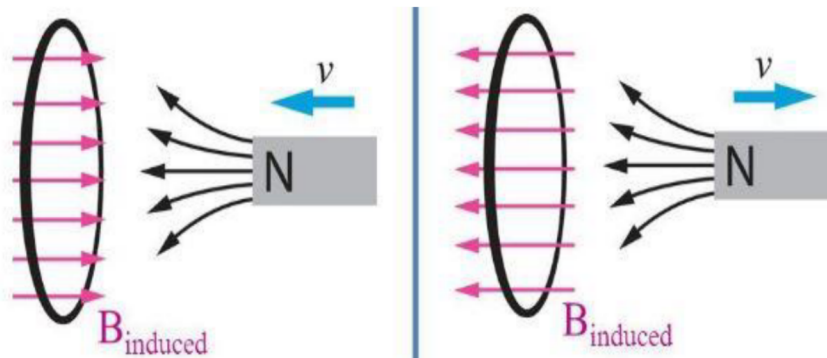


Like Oersted's law; does static magnetic field produce a current on the loop ?



Does time varying magnetic field produce a current on the loop?

Lentz's Law



Direction of current in the loop is; where induced magnetic fields opposes the incoming field through the surface.

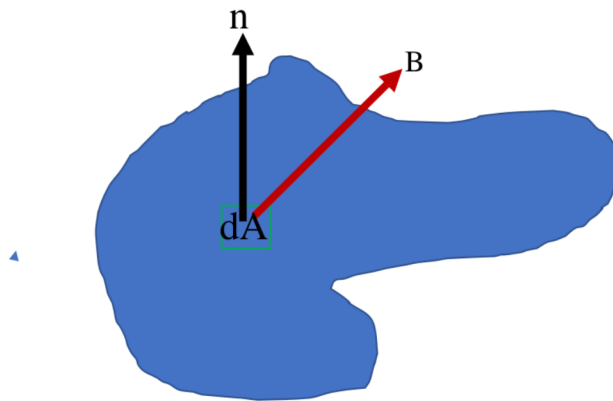
Faraday's law of Induction

Faraday's findings from experiment

- $\mathcal{E} \text{ (emf)} \propto dB/dt$
- $\mathcal{E} \text{ (emf)} \propto A \text{ (area)}$

Faraday's deduce that induced $\mathcal{E}$ (emf) is related to the magnetic flux.

Magnetic Flux



Magnetic flux; $\Phi_B = \int_s \mathbf{B} \cdot d\mathbf{A}$

Faraday's Law of Induction

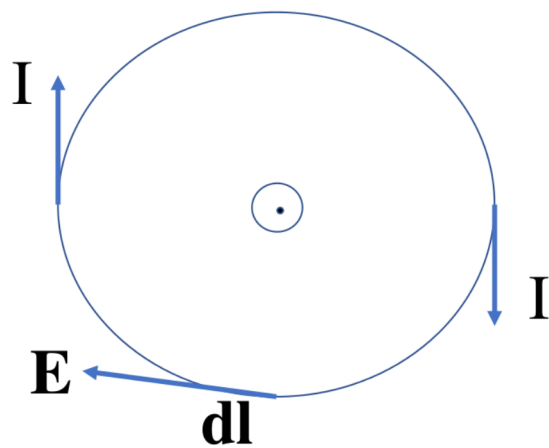
$$\mathcal{E} \text{ (emf)} = - d \phi_B / dt$$

A Changing magnetic Flux induces an EMF
and the minus sign comes from the Lenz's law.

What is emf ?

$$\mathcal{E} \text{ (emf)} = \int_C \mathbf{E} \cdot d\mathbf{l}$$

It's a “driving force” for the current.



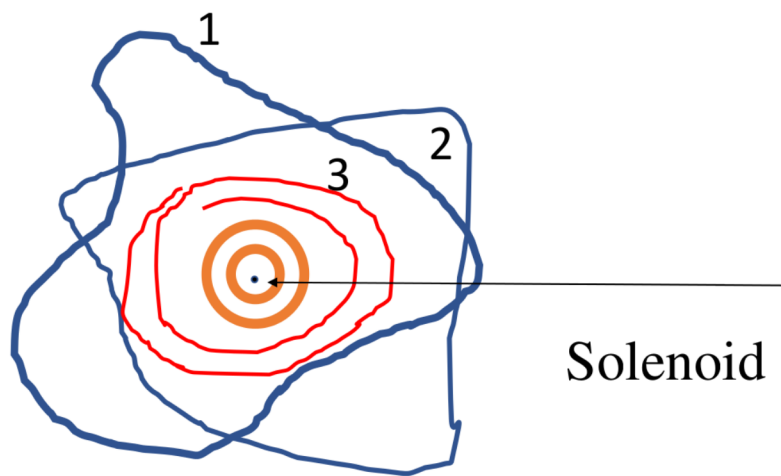
Integral form of Faraday's Law

$$\begin{aligned}\int_C \mathbf{E} \cdot d\mathbf{l} &= - \frac{d\phi_B}{dt} \\ &= - \frac{d}{dt} \left(\int_S \mathbf{B} \cdot d\mathbf{A} \right)\end{aligned}$$

Special case of Faraday's Law

$$\text{Kirchhoff's law } \int_C \mathbf{E} \cdot d\mathbf{l} = 0$$

It states that Kirchhoff's law is not valid for time varying magnetic field



$$\int_C \mathbf{E} \cdot d\mathbf{l} = -N (d\phi_B / dt)$$

- Induced emf depends only the enclosed time varying magnetic field and number (N) of enclosed loops not their shape.

Differential form of Faraday's law

$$\begin{aligned}\oint_C \mathbf{E} \cdot d\mathbf{l} &= - \frac{d\phi_B}{dt} \\ &= - \frac{d}{dt} \left(\int_s \mathbf{B} \cdot d\mathbf{A} \right)\end{aligned}$$

Applying Stoke's Theorem: $\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_s (\nabla \times \mathbf{E}) \cdot d\mathbf{A}$

Differential form of Faraday's Law: $\nabla \times \mathbf{E} = - \frac{d\mathbf{B}}{dt}$

For example any point on a close circuit : $\nabla \times \mathbf{E} = \mathbf{0}$
(Kirchhoff's law)

Applications

The most common applications of Faraday's Law of Induction are Transformer and Generator.

Thank you